CSCE 3301 - Computer Architecture Project 1

Names: Mostafa Abdelwahed
Student ID: 900243064 **Milestone:** II

Journal

1. **October 27, 2026, 2:30 pm:** Reviewed the RV32I single-cycle processor requirements and identified the main components needed for implementation.
2. **October 27, 2026, 4:15 pm:** Drew the initial CPU datapath and control diagrams before implementation, including instruction flow, register file, ALU, and memory connections.
3. **October 27, 2026, 5:40 pm:** Discussed the architecture and identified the additional components required to support the RV32I instruction set.
4. **October 28, 2026, 2:45 pm:** Implemented missing processor components and updated existing Verilog modules to support additional instructions (ALU CU, Data memory, instruction memory, etc).
5. **October 28, 2026, 4:20 pm:** Reviewed connections between the datapath components and debugged signal widths, multiplexers, and control signals.
6. **October 28, 2026, 6:00 pm:** Updated the CPU architecture diagram to reflect the implementation changes.
7. **October 29, 2026, 3:00 pm:** Developed the main CPU testbench to verify the execution of RV32I instructions.
8. **October 29, 2026, 4:30 pm:** Prepared test programs covering arithmetic, logical, memory, branch, and jump instructions.
9. **October 29, 2026, 6:10 pm:** Debugged issues identified during simulation and verified instruction execution and register results.
10. **November 1, 2026, 2:30 pm:** Integrated and tested additional processor components and reviewed the updated CPU datapath.
11. **November 1, 2026, 4:00 pm:** Debugged control signals, ALU operations, branch decisions, and jump behavior using simulation results.
12. **November 1, 2026, 5:30 pm:** Updated the processor diagram and reviewed the implementation of remaining RV32I instructions.
13. **November 2, 2026, 2:15 pm:** Ran the main CPU testbench and debugged failing instruction tests to verify correct processor operation.
14. **November 2, 2026, 4:00 pm:** Assisted in resolving integration issues between the CPU modules and verified the final datapath connections.
15. **November 2, 2026, 5:45 pm:** Contributed to project documentation, including the processor architecture, implementation details, testing methodology, and debugging results.